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BSIT 2-1

March 19, 2026

Midterm Activity # 3 - Basic ATM System

SOURCE CODE:

```
import java.util.Scanner;
public class ATMTransactionCounterA_Jose {
    public static void main(String[] args) {
        Scanner ans = new Scanner(System.in);
        double initialBal = 8000;

        System.out.println(x: " ========================== ");
        System.out.println(x: " || Welcome to A_Jose's ATM Machine || ");
        System.out.println(x: " ========================== ");

        int input = 0;
        int tCount = 0;

        while (input != 4) {
            System.out.println(x: "\n ===== ATM MENU ===== ");
            System.out.println(x: " 1. Check Balance ");
            System.out.println(x: " 2. Deposit ");
            System.out.println(x: " 3. Withdraw ");
            System.out.println(x: " 4. Exit ");
            System.out.print(s: " Enter choice: ");
            input = ans.nextInt();

            switch (input) {
                case 1:
                    tCount ++;
                    System.out.println("\n Current Balance: " + initialBal + " pesos");
                    break;
```

```
case 2:
    System.out.print(s: "\n Enter deposit amount: ");
    double depAmount = ans.nextDouble();
    initialBal += depAmount;
    tCount ++;
    System.out.println("\n Deposit successful! New Balance: " + initialBal + " pesos");
    break;

case 3:
    System.out.print(s: "\n Enter withdrawal amount: ");
    double withAmount = ans.nextDouble();
    if (withAmount > 2000) {
        System.out.println(x: "\n Error: Please don't exceed withdrawal limit of 2,000 pesos.");
    } else if (withAmount > initialBal) {
        System.out.println(x: "\n Error: Insufficient balance.");
    } else {
        initialBal -= withAmount;
        tCount ++;
        System.out.println("\n Withdrawal successfull! New Balance: " + initialBal + " pesos");
    }
    break;

case 4:
    System.out.println("\n Total transactions done: " + tCount);
    System.out.println(x: "\n ==========================================");
    System.out.println(x: "|| Thank you for using A. Jose's ATM Machine! Goodbye! ||");
    System.out.println(x: " ==========================================");
    break;
default: System.out.println(x: "\n Invalid. Please choose a number from the menu.");
}
}
}
```

OUTPUT: ==========

|| Welcome to A_Jose's ATM Machine ||

==========

===== ATM MENU =====

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter choice: 1

Current Balance: 8000.0 pesos

===== ATM MENU =====

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter Choice: 2

Enter deposit amount: 27000

Deposit successful! New Balance: 35000.0 pesos

===== ATM MENU =====

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter choice: 3

Enter withdrawal amount: 2000

Withdrawal successful! New Balance: 33000.0 pesos

===== ATM MENU =====

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Enter choice: 4

Total transactions done: 3

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!! Thank you for using A_Jose's ATM Machine! Goodbye! !!

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EXPLANATION:

The while loop keeps the ATM menu running repeatedly as long as the user hasn't chosen option 4, and once they do, the condition (input != 4) becomes false and the loop stops. The switch-case then checks the value of (input) and runs the matching case. For example, if the user enters 2, it goes to case 2 which handles the deposit, and if none of the cases match the default block tells the user their input is invalid. The if-else inside case 3 validates the withdrawal by first checking if the amount exceeds 2,000 pesos, then checking if it exceeds the current balance, and only process the withdrawal by subtracting from initialBal if both conditions pass.